

NAPLAN

State report – Year 5
December 2017

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Preface

The QCAA issues State reports on the performance of Queensland students on the National Assessment Program — Literacy and Numeracy (NAPLAN) tests. State reports provide system-level information and are publicly available. This report for Year 5 students in 2017 contains:

- the Queensland performance on each item
- the national performance on each item
- the item descriptors
- a commentary on the state results
- some recommendations for teaching.

Who should use this State report?

The NAPLAN State reports help principals, teachers and other school personnel understand, interpret and use information about student performance on NAPLAN.

School principals can use this report to provide information to the school community on aspects of the tests. This would allow professional conversations with their teachers, curriculum leaders, and department heads. Curriculum leaders can use this information to interpret the more specific information given in their school and class reports. These other reports are explained below.

Since this report is publicly available on the QCAA website, it can also inform providers of teacher training, special education services and educational research and policy.

Parents and carers can use this report to interpret the results on their child's student report. They are also able to judge how their child performed when compared with the whole population of students. The item descriptors provide them with useful information about the scope of the tests.

About the tests

The purpose of the National Assessment Program (NAP) is to collect information that governments, education authorities and schools can use to identify the important educational skills Australian students can demonstrate. As part of that program, the NAPLAN tests are administered to full cohorts of students in Years 3, 5, 7 and 9 each year. These standardised tests are sources of information about literacy and numeracy learning that can be used to inform educational policy and current educational practice.

The NAPLAN tests were developed using the nationally agreed Statements of Learning for English and Statements of Learning for Mathematics, 2005. Since 2016 however, the tests now directly relate to the Australian Curriculum. The NAPLAN tests are designed to provide a nationally comparable indication of student performance in Language conventions, Writing, Reading and Numeracy. The tests are designed to assess student understanding in the following areas:

- Language conventions: The test assesses the ability of students to independently recognise and use correct Standard Australian English grammar, punctuation and spelling in written contexts.
- Writing: The test assesses the ability of students to convey thoughts, ideas and information through the independent construction of a written text in Standard Australian English.

- **Reading:** The test assesses the ability of students independently to make meaning from written Standard Australian English texts, including those with some visual elements.
- **Numeracy:** The test assesses students' knowledge of mathematics, their ability to apply that knowledge in context independently, and their ability to reason mathematically.

Marking and scoring the tests

Marking the tests

Markers mark those test items that do not use a multiple-choice format. These markers apply nationally-agreed marking guides. There are marking guides for open-ended Reading items if any such items are included. Marking guides allow consistent and reliable judgements by markers. There are guides for the Writing test and one each for the constructed responses in Numeracy and Spelling. For some Numeracy items, students may provide a correct response in different forms. Professional officers decide on agreed scoring protocols for these items.

Calculating raw scores

The simplest calculation made in scoring the tests is the raw score — the number of questions answered correctly. All of the questions for the Language conventions, Reading and Numeracy tests are marked as either correct or incorrect. Raw scores for the Writing test are sums of the marks on each of ten criteria.

Raw scores have limited use. They enable the performance of students who have all completed the same test at the same time to be placed in a rank order, but they do not provide information about the level of difficulty of the test nor the relative differences between students.

Constructing scaled scores and bands

To make raw scores more useful, they are transferred to scores on a common scale that reflects how difficult it was to achieve each score. Each year ACARA publishes equivalence tables that allow a student's raw score to be located on the NAPLAN scale. The scale is comparable between year levels for each assessment area. An equating process is also carried out on each year's test to enable scores to be compared between successive years of testing. For example, a raw score of 20 on the Year 3 Reading test might be transformed to a scaled score of 354. This will also represent the same achievement for a student with the same scaled score in Year 5, and for a student with the same scaled score for Reading in a previous year.

Each NAPLAN scale is divided into ten bands used to report student progress.

Using NAPLAN reports to inform teaching and learning

Using scaled scores

The scaled score can compare the results of different students. Scaled scores provide a basis for measuring and comparing students' abilities across years of schooling, for example, comparing a student's result in Year 3 in 2015 and Year 5 in 2017. The scales can thus help to monitor the growth of groups of students over time. This enables the school to review and/or consolidate special programs that may have been put in place.

Principals and teachers should take care when making comparisons between small groups of students. For groups of fewer than 10 students, differences may not be reliable, particularly small differences.

Using item analysis

While the national and state reports provide the comparative data, class reports provide a school with the information that can be used to inform teaching and learning and to build capacity in schools. Analysis of the NAPLAN class data, in particular the performance on each item, will provide teachers with information about the understandings and patterns of misunderstandings of students.

Looking at the performance on the items and then analysing the error patterns allows teachers and principals to make hypotheses about why groups of students make particular errors. As mentioned below, more detailed analysis by QCAA staff is available from the QCAA website.

Steps for analysis might be as follows:

- Compare the facility rates (percentage correct) achieved by the school's students with the national and state results available in this document. Is their performance consistent?
- Look at the common errors made by their students and compare them with the common errors made in the state (only errors from Queensland students are available, and are found in the item analyses that are part of SunLANDA Online).
- Form hypotheses about why students are making these errors, e.g.
 - How did students think about this aspect of the curriculum?
 - What misunderstandings might these errors represent?
 - How might the structure of the test question have shaped the response?

Using a combination of the NAPLAN data, school data and professional judgment, teachers may then test these hypotheses to see whether they are valid or whether there is more to be thought about and investigated. Teachers can then plan lessons related to the general areas where students seem to need help. Teachers can also make judgments about teaching approaches and curriculum.

The professional conversations that are part of this process are the most effective and powerful way to use the data, as they are the vehicle for developing shared understandings.

Placing the tests in the assessment context

The results from the NAPLAN tests should be seen as only one input into a school's assessment program. Various forms of assessment are needed to inform the different stages of the teaching and learning cycle. Principals and teachers should keep in mind that NAPLAN is a pencil-and-paper, point-in-time, timed test that can only cover a few curriculum features.

The results from a school's own assessments of students should be consistent with the NAPLAN test results. If the test results are different from what was expected, consider the possible reasons. The results of the tests may indicate aspects of student performance that need further investigation within the classroom, using other forms of assessment.

An item with a low facility rate (percentage correct) may not necessarily indicate a problem in teaching and learning. It may be that this was simply a difficult item for all students in this cohort across Australia.

Other NAPLAN reports

In addition to the State reports, the following reports are produced:

SunLANDA Online

Since 2015, student data has been released on the QCAA School Portal using the SunLANDA Online interface. State schools can access the data through One School. Access to SunLANDA as application software is also still available on the QCAA website.

SunLANDA Online provides class and school information in an electronic form that permits customised spreadsheet generation by users. In addition, it shows representative samples of students' incorrect responses to constructed responses where applicable. Hyperlinks from within SunLANDA Online lead to the QCAA's test item analysis. Information on how to use this service is available at: www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda

Test Item analysis

These pdf documents contain analysis of each test item. They can be downloaded directly from the QCAA website. The analysis reproduces each item followed by expert analyses of how the item operated. It shows the distractors presented in multiple-choice items and explains students' reasoning.

School and class reports

The NAPLAN school and class reports are supplied electronically on the secure section of the QCAA website. These reports are accessible only with the school's Brief Identification Code (BIC) login and password. Individual student reports are distributed to schools as printed copies.

School reports

The QCAA issues NAPLAN School reports giving information about each school's performance. They provide a summary of year-level performance as well as performance by gender, language background and Indigenous status in the following fields:

- distribution of scaled scores
- distribution of achievement bands
- school and state means
- participation of the group.

The School report positions a school's performance within the state on a graph that is shaded to show the range of performance for the middle 60% of Queensland students together with the state mean.

Class reports

The QCAA issues NAPLAN class reports that show the performance of every student on every item. Under the name of each student is recorded the items they had correct and incorrect. They also show students' responses to constructed-response items.

The class report also gives the:

- percentage correct for each item for the class and state, and by gender
- scaled scores for each student
- performance bands for each student.

Individual student reports

The QCAA issues individual student reports to schools after the tests. Schools receive one printed report for each student to distribute to parents/carers.

ACARA Reports

As well as the Queensland reports from the QCAA, national reports are available from the website of the Australian Curriculum Assessment and Reporting Authority (ACARA). The *NAPLAN National Summary Report* and the *NAPLAN National Report* allow states and territories to place the achievement of their students in relation to their peers across the nation. This is system-level information and is publicly available.

Literacy

Writing

Stimulus (writing prompt) Years 3 & 5

YEAR 3 AND YEAR 5

Which is better?

Write a persuasive text to convince a reader that one thing is better than another.

For example, you might think that one game is more fun than another, one TV show is more enjoyable than another, or one invention is more important than another.

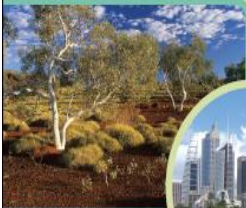
You can use your own ideas or you can use the ideas on this page.

Don't forget to give reasons for your ideas.

- **Start with an introduction.**
An introduction lets a reader know what you are going to write about.
- **Write your opinions on the topic.**
Give reasons for your opinions.
Explain your reasons.
- **Finish with a conclusion.**
A conclusion sums up your reasons so that a reader is convinced of your opinions.

Remember to:

- plan your writing
- use paragraphs to organise your ideas
- write in sentences
- choose your words carefully to convince a reader of your opinions
- pay attention to your spelling and punctuation
- check and edit your writing.



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About the task

In 2017, the NAPLAN Writing test was based on the persuasive genre. Two prompts were used, one for Years 3 & 5 and another for Years 7 & 9. The test conditions and administration remained the same as in previous years, i.e. teachers delivered the same spoken instructions and read the text aloud to students. Working independently, students had to plan, compose and edit a written response. Students were allowed five minutes to plan their script, 30 minutes to write and a further five minutes to edit and complete the task. Three pages were provided for students to write a response.

The 2017 prompt for Years 3 & 5 was titled *Which is better?* Students were asked, in the textual component of the prompt, to *convince the reader that one thing is better than another*. Some textual examples — a game, a TV show, an invention — were provided as suggestions. Additional information identified the structural components of a persuasive text and further defined these elements, e.g. *Start with an introduction. An introduction lets the reader know what you are going to write about*. Other notes were also provided in relation to the conventions associated with a writing task, e.g. *write in sentences, check and edit your writing* etc. Six photographic images accompanied by a further six adjacent comparative images were provided, e.g. a dog and a cat, fast food and healthy food.

The prompt was relatively open-ended, allowing students to base their writing on one or more of the comparative images provided, or one or more things of their own choice. Though the prompt indicated singularity (i.e. *Which **one** is better?*), nationally agreed convention permitted students to widen the comparison. Students who chose to mention a number of things did risk some loss of cohesion in their persuasive texts.

Markers for this Writing test were trained using the national persuasive writing marker training package, delivered as part of ACARA's national assessment program. Markers were recruited and trained in accordance with national protocols. For Queensland, registered teachers marked the NAPLAN Writing test. All markers applied the 10 criteria and related standards from the marking rubric. Writing test scripts were marked on screen in all states and territories. Stringent quality-control measures were applied to the marking of student scripts, including a prescribed percentage of scripts to be double-marked, and the daily application nationally of control scripts for all markers. As part of the Queensland marking operation for 2017, referee marking further ensured marker reliability. There was also provision for appeal over individual Writing test scores after the results were released. On appeal, a student's script was re-marked independently by two senior Writing test markers. A copy of the Persuasive Writing Marking Guide (2013) is available at: www.nap.edu.au/_resources/Amended_2013_Persuasive_Writing_Marking_Guide_-_With_cover.pdf

Performance

Anecdotal evidence from markers indicated that students in Years 3 and 5 were comfortable with the writing prompt, *Which is better?* The images provided were widely adopted by students as the basis for their persuasive texts, particularly the dogs/cats contrast. Cats did not generally fare well in these contrasts! A 'tour the stimulus' approach was less evident in 2017 than in some previous years. The relatively small group of students who diverged from the images provided on the prompt tended to write more challenging texts, though this was not exclusively so. In fact, students with a more sophisticated grasp of language were able to embed stylistic elements into the simple cat/dog or country/city comparison to provide a stronger sense of voice.

A closer adherence to the suggestions provided on the stimulus page appeared to be reflective of a general appropriateness for the age levels, rather than some absence of originality of ideas.

At first thought, the inclusion of a comparative element to the persuasive task may have provided greater challenge for younger writers but generally this was not evident. Students continued to

adopt fairly formulaic approaches to this persuasive text type, and whilst this broadly met the demands of the task (and rubric), it restricted a more natural flow of language and ideas. So, typically, an opening statement, such as *I firmly* (or *strongly* or even *passionately*) *believe*, would be followed by a series of body paragraphs, often introduced by the connectives *Firstly*, *Secondly*, *Thirdly* — concluding with a paragraph that provided a brief summation of arguments and ideas presented.

Students in Year 3, and to a lesser extent in Year 5, wrote in fairly straightforward sentence forms, with complex sentences common, but reflecting a limited range, e.g. *because*, *if*, etc. As far as possible, students should be encouraged to explore the full range of sentence structures, including the judicious use of short, sharp simple sentences and fragments for effect.

In terms of length of text, students in Years 3 and 5 tended to write a little more than the persuasive texts of previous years. This was encouraging. It is possible that the suitability of the prompt contributed to this trend. Obviously, students who were able to make use of a well-chosen adverb, adjective or nominal group could write with fewer words but with more demonstrable control of language.

A number of students used more varied openings to their texts to engage the reader. In some cases, this was productive.

ZZZ! Hear that? That is the sound of someone trying to play with their cat. Everybody knows that dogs are better than cats. (Year 3 student)

In other cases, the introductions appeared a little contrived, and failed to match the development of ideas (and tone) in the remainder of the text. Whilst cohesion is only one of the NAPLAN Writing criteria, the ways in which ideas, vocabulary, paragraphing and sentence formation work together is important to whole-of-text integration and reader understanding. Students should be encouraged to focus on 'big' ideas and construct their texts accordingly, ensuring consistency and sincerity of message without unnecessary gimmickry.

Relative areas of improvement from 2016 included the criteria of *Audience*, *Ideas* and *Punctuation*. The latter could possibly be because of the reduction in the use of direct speech, more associated with the narrative genre and generally more difficult. In terms of *Audience*, students wrote with greater confidence in their ability to express their personality and individuality. Since choice of *Ideas* was encouraged through the prompt, more students were able to draw on areas of interest and subject knowledge, e.g. apps, music, sport, reading genres etc.

WRITING

- Chess is Better than Mobile Games -

Minions Rush. Candy Crush. The not-so-majestic rhyming couplet extends to over a billion apps worldwide. We know that these games aren't good in the long run for us; but what can you play instead? You want a fun, challenging, competitive multiplayer game. Ladies and gentlemen, what you are looking for is chess.

The origins of chess date back to many centuries ago. It was based on many of the international battles of the time. And just like those soldiers bravely dying, using their wit to defeat the enemy so can you! Coming up with clever tricks and having to look two steps ahead at all times are some standard skills you can easily learn through a few matches of chess. You get a level of mental agility from chess you just can't get from Baby Hush or other games. As you can see, chess definitely has a one-up above games on the phone!

Secondly, chess can be played anywhere, anytime.

DO NOT WRITE OUTSIDE THE BORDER

WRITING

Chess is available in many formats, such as tiny magnetic game-boards, and nice big chess-tables. Although chess is available for download, it means that when your mum says "Phones off!" you can keep playing. Many libraries and child-care centres have small sets so you can play on your terms as well.

Finally, chess is just a whole lot more fun than games like Plumber's Brush. While many games, such as Minions Rush are just time-wasters, you'll get a smile and enjoyable time ^{from} chess. Take out your pocket chess-board, grab a mate, and start moving those pieces! Many competitions are held world-wide as well, so you can take your skills with you across the globe and play against champions. You might even win! :)

So as you can see, the all-up winner is easily chess. No good can come out of playing any mobile games. So I firmly believe that chess is better than iPhone games. And remember: take Candy Crush and put it down the Toilet Flush! :)

DO NOT WRITE OUTSIDE THE BORDER

Commentary on sample script

This surprisingly sophisticated persuasive text from a student in Year 5 extols the virtues of the traditional board game, chess, when compared to the array of mobile ‘app’ games now available to the young and not-so-young. The young writer carefully structures his text to initially engage readers, direct them to his stance, thoughtfully sift through supporting arguments involving the history, intellectual challenge, portability and competitive possibilities of playing chess, then firmly restates his position for a sense of closure, with just a touch of humour to conclude.

The writer not only demonstrates a sound knowledge of chess and convincingly promotes the advantages of playing the game, he also produces counter arguments that diminish the status of the alternative phone game apps such as ‘Candy Crush’ or ‘Plumber’s Brush’ (sic). These alternatives are *timewasters*, that *aren’t good in the long run* and that *don’t demand a level of mental agility* provided by chess.

One of the real strengths of this text is the command of appropriate language demonstrated by the writer, together with a sense of tone and writer’s voice than was naturally projected. Mature phrases — e.g. *not-so-majestic rhyming couplet; soldiers bravely dying; using their wit to defeat the enemy; level of mental agility; take your skills with you across the globe* — illustrate not only a command of vocabulary, but a capacity to elaborate and extend the meaning of that vocabulary selection. Humour based on both situation and intentional pun sustains reader interest. The final sentence — *And remember: take Candy Crush and put it down the Toilet Flush!* — is one of several examples of this feature.

In terms of the interpersonal elements, the writer actively includes the reader as the argument develops, e.g. *We know; ladies and gentlemen; so can you!; You can easily learn; As you can see*. These phrases remind the reader that they too, are aware of the position being established. These direct or inclusive addresses to the reader are interspersed with declarative statements of factual information, culminating with the definitive: *So I firmly believe*. This is the first time the personal pronoun ‘I’ is used in the text, and it definitely cements the writer’s stance.

The text scored highly in Paragraphing, with consistent use of topic sentences and paragraphs that logically sequence the development of the argument. Sentences demonstrate variety in form, length, and complexity, and flow naturally because of thematic choices, connectives and conjunctions. These range from the more straightforward — *Secondly, Finally, So as you can see* — to interesting theme constructions like *You want a fun, challenging competitive multiplayer game; Coming up with clever tricks; No good can come out of playing*.

Punctuation, with just a minor slip or two, helps pace and direct the reader through the text and spelling is flawless. Overall, the text provides a quality exemplar of this text type.

Language conventions

Spelling

Results and item descriptions

The percentage columns give the facility rate (percentage correct). These results are based on provisional data.

Item	Answer	Qld %	Aust %	Description
Proofreading — Error identified				
1	mother	89.6	91.22	Correctly spells a two-syllable word with a short vowel sound.
2	soil	75.38	75.5	Correctly spells a one-syllable word with the vowel digraph -oi-.
3	always	93.57	93.38	Correctly spells a two-syllable word with an initial letter pattern al-.
4	humming	86.29	86.19	Correctly spells a two-syllable word with the suffix -ing requiring doubling of a final consonant.
5	loaded	70.94	71.89	Correctly spells a two-syllable word with the vowel digraph -oa-.
6	raised	70.33	72.06	Correctly spells a one-syllable word with the vowel digraph -ai-.
7	creaks	65.83	65.49	Correctly spells a one-syllable homophone with the vowel digraph -ea-.
8	dolphins	55.14	58	Correctly spells a two-syllable word with the consonant digraph -ph-.
9	figure	31.45	35.15	Correctly spells a two-syllable word with the ending -ure.
10	whistle	52.78	53.09	Correctly spells a two-syllable word with the initial consonant digraph wh-.
11	politely	31.77	34.1	Correctly spells a three-syllable word with the -ly suffix.
12	columns	18	18.56	Correctly spells a two-syllable plural word with a silent letter -n-.
Proofreading — Error not identified				
13	native	84.04	85.28	Identifies an error, then correctly spells a two-syllable word ending in -ve.
14	warrior	67.4	69.89	Identifies an error, then correctly spells a three-syllable word with -rr- at a syllable juncture.
15	equal	55.1	55.98	Identifies an error, then correctly spells a two-syllable word ending in an unstressed syllable -al.

16	interesting	47.13	50.19	Identifies an error, then correctly spells a multisyllable word with an elided syllable.
17	women	40.84	44.77	Identifies an error, then correctly spells a two-syllable irregular plural word.
18	thread	39.11	40.68	Identifies an error, then correctly spells a one-syllable word with the vowel digraph -ea-.
19	laughter	53.9	54.51	Identifies an error, then correctly spells a two-syllable word with the digraphs -au- and -gh-.
20	relief	38.57	40.19	Identifies an error, then correctly spells a two-syllable word with the vowel digraph -ie-.
21	knives	33.14	32.43	Identifies an error, then correctly spells a one-syllable plural word ending in -ves.
22	tricycle	31.39	34.14	Identifies an error, then correctly spells a three-syllable word with a prefix tri- added to a base word with the short vowel sound /I/ represented by -y-.
23	deliberate	20.85	23.9	Identifies an error, then correctly spells a multisyllable word with an elided syllable.
24	representative	7.14	10.05	Identifies an error, then correctly spells a multisyllable word with a schwa sound.
25	exercises	11.57	12.05	Identifies an error, then correctly spells a multisyllable word with -x- representing two sound values.

About the test

The 2017 Year 5 test involved the following spelling features:

Short vowels and the schwa sound: *mother*, *representative*

elided vowels: *interesting*, *deliberate* (and also *gathering* as a distractor word)

unstressed final syllables: *equal*, *bucket*, *figure*, *warrior*

complex vowel spellings: *soil*, *creaks*, *loaded*, *raised*, *women*, *thread*, *tricycle*, *relief*, *laughter*

complex consonants: *dolphins*, *exercises*

syllable juncture consonants: *dol-phins*, *war-rior*

inflection: *humming*, *knives*

affixes: *native*, *always*, *politely*

silent letters: *laughter*, *columns*, *whistle*, *knives*

Performance

Queensland Year 5 students in 2017 performed a little better on spelling than did those in 2016. They performed better than the national average on four words: *knives*, *creaks*, *humming* and *always*. Queensland students performed least well on the words *equal*, *representative*, *deliberate*, *figure* and *dolphins*.

Students attempted most of the questions. However, around 10% of students omitted giving any response regarding the difficult words, *exercises*, *representative*, *deliberate* and *knives*. When

students omit answering questions, they do not provide evidence about their spelling knowledge on those tasks.

Girls performed better than boys on all items except *warrior*. The difference was often not great but was greater than 10% on the words *dolphins*, *laughter*, *thread*, *equal* and *whistle*. Boys were also more likely than girls to omit a response.

Student error patterns reveal that many over-rely on ‘sounding out’ spelling strategies, especially when confronted with less familiar words, e.g. *representative*, *gathering* and *exercises*.

The proofreading format used in NAPLAN may lead students to show different spelling behaviour than when they spell to dictation or when they generate spelling in the course of composing their own sentences. Teachers should judge the results in this context.

See the full reports on each item in the SunLANDA Online program and as PDF documents at <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>

Implications for teaching

Please refer to the extensive discussion of the teaching of spelling in the NAPLAN State Report for last year, 2016. State reports of previous years are available on the QCAA website.

Testwiseness

Attempt all questions: Students need to pace themselves through the test. They should make an attempt and move on, leaving enough time to come back to the doubtful answers.

Read the word in context: The first section of the test identifies the misspelled target word with a circle. But the circled word alone is sometimes misleading. Students must read the word within the context sentence. For example, some students saw ‘soyl’ and decided the target word was ‘soul’, but the sentence shows that the real target was *soil*.

Gender

Boys’ performance over a number of years of testing suggests that they take pride in their literacy knowledge if they perceive it as associated with boys. Year 5 boys spelled *warrior* better than girls did. In Year 3, boys spelled the words *fighting* and *tricycle* as well as girls did. Year 7 boys spelled the word *recruit* and they understood the concept of a *command* better than girls did. Lessons on spelling could be built around words such as these that boys are motivated to spell properly.

Beyond ‘sounding-out’

Phonemic awareness: Students have to learn how the sounds match letters alphabetically in many single-syllable words. Students need phonemic awareness or the ability to hear all the sounds inside a word and they need alphabetic knowledge of the sounds of each letter. Next, patterns containing more than one letter are learned, such as the digraph oi for the diphthong sound and the CVC pattern whereby a long vowel sound is shown by a final e.

Morphemic awareness: Students then learn about the chunks of letters with grammatical meanings. These chunks include plural inflections (*knives*) tense inflections (*humming*), prefixes (*always*) and suffixes (*native*).

Full reports on student performance are in the SunLANDA Online program and as PDF documents at <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>

Grammar and punctuation

Results and item descriptions

The percentage columns give the facility rate (percentage correct). These results are based on provisional data.

Item	Answer	Qld %	Aust %	Description
26	C	89.01	88.82	Identifies the correct contraction for <i>could not</i> .
27	B	93.85	94.34	Identifies the correct modal operator in a complex sentence.
28	A	87.65	88.01	Identifies the correct dummy subject for a noun clause in a complex sentence.
29	A	83.01	83.51	Identifies the correct auxiliary verb for past perfect tense in a complex sentence.
30	B	88.84	87.68	Identifies the action verb in a complex sentence.
31	B	91.02	90.21	Identifies an adverb in a complex sentence.
32	C	42.6	45.96	Identifies whether a noun is singular or plural.
33	A	60.67	62.94	Identifies the correct placement of clause commas in a complex sentence.
34	C	75.48	76.7	Identifies a place noun in a simple sentence.
35	C	60.88	60.36	Identifies a word used as a verb in a complex sentence.
36	BE	54.31	56.57	Identifies two subordinate clauses that meaningfully complete a complex sentence.
37	D	66.38	67.38	Identifies the correct punctuation of a list in a simple sentence.
38	D	54.81	54.84	Identifies a simple sentence requiring quotation marks.
39	A	43.99	42.84	Identifies the sentence with the greatest degree of certainty.
40	D	35.6	36.34	Identifies the correct relative pronoun to complete a complex sentence.
41	A	44.76	45.56	Identifies the correct placement of quotation marks for direct speech with internal attribution.
42	A	35.83	36.35	Identifies the last event in a series in a complex sentence.
43	B	31.93	34.52	Identifies the subject of a compound sentence.
44	B	32.75	32.37	Identifies the correct use of capital letters for geographical names and features.
45	AB	27.59	29.04	Identifies the correct placement of clause commas in a complex sentence.

46	D	28.3	26.84	Identifies that a main clause contains subject and verb.
47	B	37.81	36.81	Identifies a sentence containing reported speech.
48	D	36.86	38.93	Recognises the use of connectives to link sentences.
49	C	27.66	29.08	Identifies the reason for the use of an apostrophe in a simple sentence.
50	D	26.56	26.4	Identifies the correct pronoun to match the noun it references in a complex sentence.

About the test

The NAPLAN Language conventions items test sentence-level, clause-level and word-level skills. The test does not cover the curriculum. Instead, it tells how a large number of students perform on a small range of tasks. Standardised tests can, however, suggest broad trends across a cohort. At the level of individual students, NAPLAN results can supplement classroom assessments and guide teachers to important points of grammar and punctuation that need revisiting.

The 2017 Year 5 test involved the following features:

Cohesion: Students had to follow the causal and sequential links between ideas created by conjunctions and verb tense (Items 28, 29, 36, 42, 43 and 48).

Sentence structure: Students had to identify, by both name and function, the subject of a sentence (see Items 43 and 28) and the verb in a clause (Items 30, 35 and 46). They had to make pronouns and verbs agree with the subject (Items 46 and 50).

Adverbs: Students had to identify adverbs by name and function (Items 31 and 46).

Modality: Item 39 involves the modal auxiliaries *must* and *need to*. Item 27 was about the modal adjective, *possible*.

Commas: Students had to identify correct clause commas in Items 33 and 45, and to identify correct list commas in item 37.

Direct speech: Students had to know the punctuation of and the typical sentence structures that distinguish indirect, reported speech from direct, quoted speech (Items 38, 41 and 47).

Apostrophes: Students had to know that *couldn't*, *friend's* and *who's* are verb contractions (Items 26, 40 and 49). They had to recognise a possessive plural apostrophe (Item 32).

For information about the full range of grammar knowledge that Year 5 students should have, refer to the Australian Curriculum English.

Performance

Although Queensland Year 5 students performed a little below the national average on most questions, they did a little better on Items 30 and 46, which both asked students to identify the verb in the sentence. They were also better on Item 39, which was about modality. A similar good performance on a question involving modality was observed on the Year 9 test. These above-average results suggest that targeted teaching of the names of the parts of speech and of modality have had an impact in Queensland.

However, Queensland Year 5 students performed noticeably below the national average on Item 43 (*Which word is the subject of this sentence?*) It would seem that students' knowledge of the names of the parts of speech is not matched by their knowledge of the names of the components

of a sentence. Queensland students also found Item 32, about possessive plural apostrophes, especially difficult.

As was the case in 2016, a question about capitalising proper nouns proved difficult.

Girls outperformed boys a little on most questions. The performance gap reached about 10% on tasks that required persistence and attention to detail, namely:

- Items 33 and 37, about clause and list commas
- Item 38, about identifying where direct speech punctuation is needed
- Item 48, about rephrasing information into a complex sentence.

Implications for teaching

Grammar and punctuation are not separate areas but components of reading and writing. Although NAPLAN tests grammar and punctuation at the level of single sentences, this is not the way to teach or assess these skills in the classroom. Rather, teach how a sentence fits into a wider text as this will influence choices about the sentence's pronouns, its verb tense, its order of components (subject, verb and object) and its elaborations.

Metalanguage

Before teachers and students can talk about the more engaging challenges of constructing a rich and coherent text, they must be able to identify and name the building blocks of sentences and know how to use them.

Proofreading

Students need practice in proofreading so that they are able to proofread their work and recognise errors in grammar, weaknesses in style or inappropriate grammar for a specific genre. It is also important to expose students to exemplary texts and point out how a sentence can be crafted, balanced, given pace and rhythm and contribute to the tone or meaning of the whole text.

Commas

The commas that separate a subordinate clause from the main clause obey simple conventions. Students should learn about this as they learn how and why the subordinate clause can be placed in different positions in relation to the main clause.

Direct speech

The tense sequence and the use of first and third person pronouns for reported and direct speech follows conventions that students must learn (e.g. She said '**I will** attend' / She said that **she would** attend).

Apostrophes

Many students struggle to understand the concepts of contraction, possession and plurality and how they relate to spelling and apostrophes. It is a great gift for a student to know the rules governing contractions and possession clearly in the early years.

Testwiseness

Students would benefit from being taught techniques for maintaining persistence and being systematic.

Although NAPLAN is a test of written, standard, Australian English, it often uses example sentences that are from informal, spoken situations. Familiarity with diverse types of texts may help students to be more confident in viewing the NAPLAN items. Guide students through notable grammar and punctuation in a wide selection of reading materials, including texts that are challenging and divergent in form.

QCAA resources

Please refer to SunLANDA Online, which is available to schools via the QCAA Portal on the QCAA website through the school BIC and password. The SunLANDA Online program displays the school's results but also links to detailed analysis of every item on the NAPLAN test. The analyses include Australian Curriculum links, language resource texts and other QCAA materials. The item analysis is also available collected into PDF format on the NAPLAN pages of the QCAA website.

Reading

Results and item descriptions

The percentage columns give the proportion of correct answers (facility rates). These results are based on provisional data.

Item	Answer	Qld %	Aust %	Description
<i>Wanted</i>				
1	B	98.51	98.61	Locates directly stated information in a simple poster.
2	A	97.75	97.83	Interprets directly stated information in a simple poster.
3	D	94.11	93.37	Locates directly stated information in a simple poster.
4	A	97.69	97.73	Locates directly stated information in a simple poster.
5	C	97.31	97.3	Locates directly stated information in a simple poster.
<i>Bubbles</i>				
6	D	86.96	86.04	Identifies the first-person narrator in a poem.
7	B	60.97	60.19	Interprets the meaning of vocabulary in a poem.
8	D	71.74	69.18	Interprets the meaning of a phrase in a poem.
9	C	91.24	91.49	Interprets the meaning of a description in a poem.
10	A	90.92	91.22	Interprets the meaning of a sentence from descriptive language in a poem.
11	B	93.05	93.12	Infers the meaning of a description from context in a poem.
12	A	72.03	72.48	Analyses information across stanzas in a poem.
<i>Beached</i>				
13	C	68.38	68.12	Interprets directly stated information in a narrative.
14	A	58.71	58.51	Identifies the reason for a character's actions in a narrative.
15	A	76.33	76.59	Links information across sentences in a narrative.
16	D	28.27	28.61	Identifies a change in a character's perspective in a narrative.
17	C	75.26	75.95	Locates directly stated information in a narrative.
18	B	72.57	73.37	Interprets directly stated information in a narrative.
19	B	50.79	51.86	Infers the reason for a character's actions in a narrative.
<i>Something to write on</i>				

20	B	78.86	79.44	Identifies the genre of an information report.
21	A	55.88	56.71	Locates directly stated information in an information report.
22	C	50.9	51.88	Infers a key idea from a paragraph in an information report.
23	C	74.22	73.89	Infers a key idea in an information report.
24	D	51.41	51.31	Links directly stated information across sentences in an information report.
25	B	41.54	41.36	Interprets the meaning of a phrase in an information report.
<i>Fit to reveal fitness?</i>				
26	D	43.11	44.82	Identifies the main idea of a persuasive article.
27	B	65.8	65.31	Identifies the main purpose of a sentence in a persuasive article.
28	C	33.21	33.74	Interprets the meaning of complex language in a persuasive article.
29	D	26.51	28.47	Interprets the meaning of vocabulary in context in a persuasive article.
30	A	41.99	44.78	Locates directly stated information embedded in a paragraph in a persuasive article.
31	C	43.55	44.71	Identifies the purpose of evidence in a persuasive article.
32	D	29.17	30.56	Identifies the purpose of italics in a persuasive article.
<i>The Burrunan dolphin</i>				
33	D	58.77	62.11	Identifies the purpose of a paragraph in a detailed informative article.
34	C	53.67	53.93	Interprets the meaning of vocabulary in context in a detailed informative article.
35	D	73.23	74.61	Identifies the purpose of a paragraph in a detailed informative article.
36	B	29	30.52	Analyses information across the text in a detailed informative article.
37	A	51.84	54.94	Interprets the meaning of vocabulary from context in a detailed informative article.

About the test

In 2017, the Year 5 Reading test consisted of 37 items based on six reading magazine units with the following text-types or genres:

- information — *Something to write on* and *The Burrunan dolphin*
- persuasion — *Fit to reveal fitness*
- poetry — *Bubbles*
- imaginative narrative — *Beached*
- poster — *Wanted*.

While facility rates are expected to gradually decrease from the beginning to the end of the test, in this year's reading test it was the second last unit, *Fit to reveal fitness*, that proved most challenging for students. As this text also appeared in the Year 7 Reading test it could be deemed a more difficult text. However, the prevalence of items that required students to identify the purpose of elements of the text would have also contributed to this unit's difficulty.

Teachers can view school-specific performance information through the QCAA's SunLANDA Online program. SunLANDA Online is available on-line through the QCAA Portal on the QCAA home page. State schools can also access this content through *OneSchool*. SunLANDA Online displays the performance of classes, subgroups, and individuals within the school and compares the school's performance with that of the state and nation. Most importantly, hyperlinked to each item are the analyses and teaching ideas to help teachers and students with this type of question.

Performance

It was pleasing to see that 94.5% of Queensland students performed at or above the national minimum standard, compared to 94.0% nationally. The first unit, *Wanted*, was a simple text that placed very few demands on students, who were mainly required to locate information that was explicitly stated in the text. The next unit, *Bubbles*, required students to interpret a poem and make several inferences regarding vocabulary, narrative perspective and gaps in the text. Despite the prevalence of these higher-order items, the facility rates for this unit were high. The complexity of the texts increased with the narrative *Beached*, which also appeared in the Year 7 Reading test. Item 16 from this unit proved to be one of the most challenging items on the test with the second lowest facility rate (28.27%). This item required students to interpret the text to establish how the characters' feelings change at a particular point in the narrative. This low facility rate might be attributable to the students' lack of familiarity with the word *marvel*, a key word in the text.

Students generally had little difficulty identifying *Something to write on* as a historical report, but had less success when required to make inferences and interpretations (Items 22 and 25). *Fit to reveal fitness?* had medium to low facility rates with Item 29 proving to be the most difficult item on the test (26.51%). To a large extent, this item relied on the student's familiarity with the word *innovative*. However, students who were not familiar with this word could have worked out its meaning from the context in which it is used. Item 32 required students to identify how italics functioned in the text. Student responses were spread fairly evenly across the options, which all offered feasible explanations of how italics might function in texts. However, students who answered incorrectly did not consider how italics were operating in this particular text. Facility rates for the last unit, *The Burrunan dolphin*, were generally higher than the previous unit. Item 36 required students to locate specific information that appears in the text and discount other information that does not appear in the text. The low facility rate (29%) for this item, which is essentially a low-level literal recall question, might be attributable to the fact that students needed to read the entire text to make sure that particular pieces of information were missing from the text.

Generally questions that involved purpose, tone and character responses had lower facility rates than literal and lower-order inferential questions. This is because they required higher-order

reasoning and comprehension, i.e. students had to form an understanding of the whole text as well as pay attention to subtle clues in the text which help them make the inferences.

Implications for teaching

This year as usual, Year 5 students demonstrated a high capacity to answer literal (recall and translation) type questions. But items requiring students to make higher order inferences continued to be challenging, suggesting that students need to develop strategies to help them make inferences as they read, i.e. to draw conclusions about the unknown based on the known.

As a general note, all questions involving purpose, main idea, theme or tone of the text (in whole or part) challenge students because they have to understand the whole of the text to answer the question. The big challenge for teachers is to get students to read a variety of texts, annotate them in the classroom and discuss the ideas in the texts in groups so that they can see how all the parts of the text contribute towards the meaning of the whole. This is the time to discuss patterns in the text (e.g. cause and effect, contrast, comparison), identify connections between ideas in the text as well as the two or three main parts of the text and how the parts contribute to the overall meaning. All of this should occur before students begin a close study of the text. Students will handle the distractors in the questions much better if they are clear about the subject matter and the purpose of the text **before** they proceed to the questions. The QCAA website provides teacher advice on specific strategies for annotating and discussing challenging texts with a focus on helping students make inferences.

Teachers need to encourage students to read for pleasure and recreation to extend their knowledge of themselves and the world around them. Reading develops empathy for characters and people in difficult situations. Students also need to be able to confidently participate in a close study of a text, to check for fallacies and persuasive techniques, and to draw attention to emotive language and literary techniques. World citizens need to be discerning and capable readers and confident speakers and writers about those texts.

The complexity of the reading process becomes apparent when students discuss texts and share how they arrive at their personal understanding of the text. Students need to experience using extracts in which not everything in the text is clear (e.g. sometimes a character's motivation is not clear), and students should not panic if the subject matter is not completely accessible).

Teachers are the facilitators of this process of annotating and discussing texts. They are not the leaders. Teachers should focus on:

- modelling a love of books and reading
- finding authentic texts which appeal to children of that age
- promoting higher-order questioning of texts (both set texts for special study and unseen texts for close study)
- reading aloud to students to promote reading for pleasure
- talking about texts and authors respectfully and disagreeing with each other about their interpretations appropriately (Where is your evidence for that?)
- developing an awareness of how the parts of the text combine to create a whole through both semantic (links between the ideas) and syntactic (grammatical links) cohesion
- encouraging students to make inferences as they read (i.e. an informed guess backed by evidence from the text)
- encouraging students to see connections between the text and their own knowledge and experience, between different things within the text and between this text and other texts in a

similar genre or on similar subject matter. This will assume increasing importance as students approach secondary schooling

- encouraging students to be active readers and make connections between the text and their own knowledge, experience and feelings.

QCAA resources

SunLANDA Online displays item analyses along with data on the performance of classes, subgroups, and individuals within a school and compares the school's performance with that of the state and nation.

QCAA 2015, *Beyond NAPLAN: How to read challenging texts*, Beyond NAPLAN series,
http://www.qcaa.qld.edu.au/downloads/p_10/naplan_read_challenging_texts.pdf

Numeracy

Results and item descriptions

The numeracy strands are abbreviated as follows: *Number and Algebra* (NA); *Measurement and Geometry* (MG); *Statistics and Probability* (SP). All items are worth one score point.

The percentage columns give facility rates (percentage correct). These results are based on provisional data.

Item	Strand	Answer	Qld%	Aust%	Description
1	NA	C	97	97.17	Identifies an addition expression that matches a given value.
2	SP	B	96.97	97.02	Identifies which event is most likely to occur.
3	NA	B	96.08	96.15	Determines half of the number of items in a display.
4	SP	B	94.37	94.54	Interprets information from a two-way table.
5	MG	D	88.78	89.86	Converts weeks to days.
6	MG	A	89.33	89.29	Identifies the effect of a one-step flip.
7	NA	A	91.77	92.25	Identifies collections of coins with the same value.
8	NA	A	84.47	85.58	Solves a problem using the division facts of five.
9	NA	D	77.28	77.77	Uses an addition pattern to determine the sum of the next two terms in a sequence.
10	NA	D	75.95	76.27	Adds decimals in context.
11	NA	12	78.86	80.31	Uses a pattern to identify a missing number in a table.
12	NA	B	88.21	85.85	Counts by halves including mixed numerals.
13	SP	C	71.23	71.44	Identifies a question that would result in the given column graph.
14	NA	34	73.89	74.48	Solves a subtraction problem using basic facts and place value.
15	NA	12063; 4302	58.63	61.42	Reproduces four- and five-digit numbers in their numerical representations given words.
16	MG	C	70.44	70.09	Compares the dimensions of a rug to the dimensions of a room.
17	MG	A	69.57	71.31	Chooses an appropriate unit of measurement for capacity.
18	MG	F	60.76	61.12	Uses directional language to describe a route.
19	NA	C	64.21	65.04	Uses place value to regroup numbers to solve an addition problem.

20	NA	B	73.09	70.45	Identifies the decimal form of the fraction 5/10.
21	NA	D	54.94	56.38	Identifies the correct number sentence to model a 'how many more' problem.
22	MG	3;4	54.6	56.74	Compares features of different three-dimensional models.
23	MG	D	50.97	53.48	Identifies a composite shape with a line of symmetry described in words.
24	NA	120	42.94	45.87	Solves a multi-step multiplication problem.
25	NA	B	60.99	60.92	Identifies the number less than 0.3 and greater than 0.16.
26	NA	C	39.8	39.3	Identifies equivalent fractions used in context.
27	MG	B	57.5	59.93	Compares the capacity of objects using informal units.
28	NA	B	47.46	49.47	Solves a division problem involving the interpretation of the remainder.
29	SP	A	70.11	69.38	Selects a picture graph to represent given data where one picture represents 3 pieces of data.
30	NA	D	48.88	51.58	Identifies a word problem for a given multiplication number sentence.
31	SP	D	67.78	69.6	Identifies an error in a column graph.
32	MG	A	49.05	52.59	Compares volumes of solid objects.
33	NA	D	40.54	42.92	Uses reasoning, addition and subtraction to solve a word problem.
34	SP	C	48.74	51.49	Recognises that probability ranges between 0 and 1.
35	NA	100	26.2	29.66	Multiplies a decimal by a whole number to solve a problem.
36	MG	C	29.54	29.95	Compares lengths expressed in different units and identifies the longest.
37	NA	AC	38.12	37.5	Uses the inverse relationship between addition and subtraction to identify the number sentences to solve a word problem.
38	NA	50	22.98	22.27	Solves a problem involving comparing objects by length using metres and parts of metres.
39	NA	B	27.37	24.47	Identifies a decimal with the given digit in the thousandths.
40	MG	4:20	23.89	27.64	Calculates the duration of time in a multi-step problem.
41	MG	2050	10.86	7.81	Converts from kilometres to metres to compare distances.

42	NA	25	6.56	8.02	Solves a multi-step problem involving the addition and subtraction of numbers in a number sequence.
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About the test

As was the case in 2016, The Year 5 numeracy test covered concepts and skills from across the Australian Curriculum strands of Number and Algebra (24 items), Measurement and Geometry (12 items), and Statistics and Probability (6 items). In 10 of the items students were required to construct a response, as opposed to making a multiple-choice selection. This number of constructed responses was higher than in previous years in Year 5. On these types of items, Queensland students were less successful than the national cohort in 80% of cases.

Performance

Overall though, Queensland students performed comparably with the national cohort, with the performance on 20 items at or above the national mean percentiles out of the 42-item test. This year, 96% of Queensland Year 5 students scored at or above the national minimum standard, compared to 95.4% of Australian students overall. There were also several items where the Queensland facility rate was higher than the national rate with most of these in the Number and Algebra strand. Another positive feature of the performance of Queensland students was that in 4 of the final 6 items on the paper (items generally constituting the most difficult subset), Queensland students outperformed the national cohort average. In Item 39, a difficult item that required students to identify a decimal with a given digit in the thousandths, the mean of Queensland students was almost 3% higher than the national mean.

Following the gradient of difficulty used in standardised tests, facility rates lowered as the test proceeded, with 98% of students correct for Item 1, 61% for Item 25, down to 6% for Item 42.

Particular items of interest were:

- Item 12: An item requiring students to count by halves (Qld almost 3% higher than national)
- Item 20: An item requiring conversion of fraction to decimal (Qld almost 3% higher than national)
- Item 39: An item requiring students to identify a decimal with a given digit in the thousandths (QLD almost 3% higher than national)
- Item 41: An item requiring students to convert kilometres to metres (Qld 3% higher than national)
- Item 15: An item requiring students to convert word representations to numbers (Qld almost 3% lower than national)
- Item 32: An item comparing relative volumes of prisms (Qld 3% lower than national)
- Item 40: An item requiring students to calculate duration of time in a multi-step problem (Qld almost 4% lower than national).

In previous NAPLAN Numeracy tests, it has been relatively common for students to omit responses in the final subset of items, i.e. those items that are more challenging. In 2017, items with the highest omit rates were Items 5, 33, 35, 40, 41, 42 and 38. In item 38, 8% of students failed to provide a response. This quite difficult 'number' item (a pre-algebraic form) required students to determine the height of an object by comparing it with another object's height. Additionally, students had to convert metre measure to centimetre measure.

More challenging items of this type provide students with opportunities to apply more developed skills and strategies that differentiate among the proficient students. Queensland students slightly underperformed (compared to the national cohort) on items that required a constructed response. Some students may lack the confidence to commit to a constructed response of their own making. A review of incorrect responses shows that many students may have answered only part of the given problem.

Implications for teaching

Testwiseness

Students should expect constructed-response items to be included on the test as well as multiple-choice questions. They need to have problem-solving strategies ready. Students can develop their mathematical confidence by solving complicated tasks requiring them to construct their response independently. A benefit could be improved test performance on standardised test items requiring constructed responses.

Problem solving

Successive *NAPLAN State Report* commentaries have reiterated the importance of problem-solving strategies, particularly at Year 5 level and beyond. Problem solving involves a plan or process and using a variety of methods, either learned or reasoned, in a logical manner to find a solution. Students require opportunities to make their own decisions about how to solve a problem. Teachers can pose maths problems derived from different curriculum areas.

Many teachers have made use of the four-step problem-solving model from the book, *How to solve it*, by George Polya (Princeton University Press, 1949). The four steps are:

1. Understand the problem — One of the challenges in teaching problem solving to young children is to ensure they understand the nature of the problem. Recognising what is actually being asked in a task involving a numerical solution, whether it is a test item or a classroom task or a real-life problem, is often the key to strategising correctly. Students should look for what information is given and what is the unknown to be discovered. It may help students to draw a diagram. They should also introduce suitable notation related to the problem.
2. Devise a plan — Find the connection between the known data and the unknown. Sketch out a plan of the solution.
3. Carry out the plan — Check each step as you go. Can you prove that each step is correct?
4. Look back — Examine the solution obtained. Check its reasonableness.

While all steps are important, the ‘checking for reasonableness’ is particularly beneficial in test situations, especially when students are asked to construct a response.

Tables, graphs, diagrams

Of the 42 items on the 2017 Year 5 Numeracy paper, 19 items required interpretation of a diagram, table or graph or involved some graphical image. It is important that students have skills to interpret graphics, gain wide mathematical content knowledge and use strategies to solve a range of problems. Strategies include visualisation, spatial recognition, line measurement and estimation.

Students need to recognise that differing ways in which data can be represented are important elements of numeracy. Teacher-led class discussions about mathematical diagrams will help students develop the skills and experience required to interpret them and to create diagrams to solve word problems.

QCAA resource

Please refer to SunLANDA Online for a detailed analysis of individual test items, including teaching ideas designed to assist with the development of the understanding and skills required by each item. SunLANDA Online is available to all schools on the *QCAA Portal* link on the QCAA website. Additionally, SunLANDA materials are available to State schools through OneSchool.

QSA, 2005, *Mathematics: About space*.

https://www.qcaa.qld.edu.au/downloads/p_10/kla_maths_info_space.pdf